AXI4-Stream to PCIe 6.0 Bridge

Round 2: Integration, Verification, Optimisation and Physical Implementation

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Abstract

This report documents the transition from the independently verified RTL modules of Round 1 to an integrated, regression-tested AXI4-Stream-to-PCIe 6.0 bridge in Round 2. The implementation targets device-to-host DMA for SmartNIC packet ingestion and ML accelerator result streaming. A 512-bit transaction datapath, 256-byte Flit construction, replay/CRC/FEC mechanisms, credit-aware weighted arbitration, and a parameterized PHY interface were evaluated in simulation and synthesized for a Xilinx xcu250 FPGA. The team's final implementation ledger reports 255 MHz on the PCIe clock with timing met, 6.17 GB/s for simulated ML tensor traffic and 4.84 million packets/s for simulated SmartNIC IMIX traffic. The independently supplied Vivado timing and resource reports additionally provide a specific 250 MHz constrained run, with positive worst-negative slack of 0.349 ns and 57,184 LUTs. The SKY130 interconnect block produced a DRC/LVS/antenna-clean GDSII; complete-bridge physical-design closure is not established by the supplied evidence. Values attributed to different runs are explicitly distinguished below.

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1 Round 1 Status and Transition to Round 2

Round 1 established the three-layer RTL partition and verified the constituent modules using directed testbenches. The interconnect carried 128/128 checked beats with zero errors; the protocol-to-PHY Gray/PAM4 chain passed loopback and 4/4 cocotb tests; and the link-layer core had functional Flit framing and buffering. An initial `system_top.v` demonstrated end-to-end *connectivity*, including a simulated PHY loopback. However, it did not yet prove fully correct device-to-host packet delivery: packetizer/address translation and other tested transaction components were incompletely wired, and the link–PHY boundary lacked ready/backpressure. Round 2 therefore prioritized interface review, missing RTL integration and end-to-end correctness before physical optimization.

The official problem statement specifies RTL development, simulation, verification of transactions and reset/error cases for the qualifier; the final round adds optimization, workload evaluation, synthesis, open-source physical design and GDSII generation. The implementation here intentionally focuses on two AXI4-Stream device-to-host producers rather than a fully bidirectional AXI4 memory-mapped master/slave bridge. This is an explicitly documented scope decision, not a claim of complete bidirectional AXI4 functionality.

2 Round 2 Objectives and Scope

The two workloads are (i) SmartNIC frames of variable size with high packet rates and (ii) long ML-inference output tensors requiring sustained bulk throughput. Both share the PCIe egress datapath but retain independent per-channel session counters and buffer-base addresses. Required functions include clock crossing, flow control, address generation, TLP packing and 4 KB splits, PCIe 6-style Flit construction, replay, error detection/correction and host-control sideband. The digital PHY is modeled through logical/Gray/PAM4 lanes, without claiming a fabricated analog SerDes or live PCIe compliance qualification.

3 Integrated Architecture

3.1 Top-level Vector Block Diagram

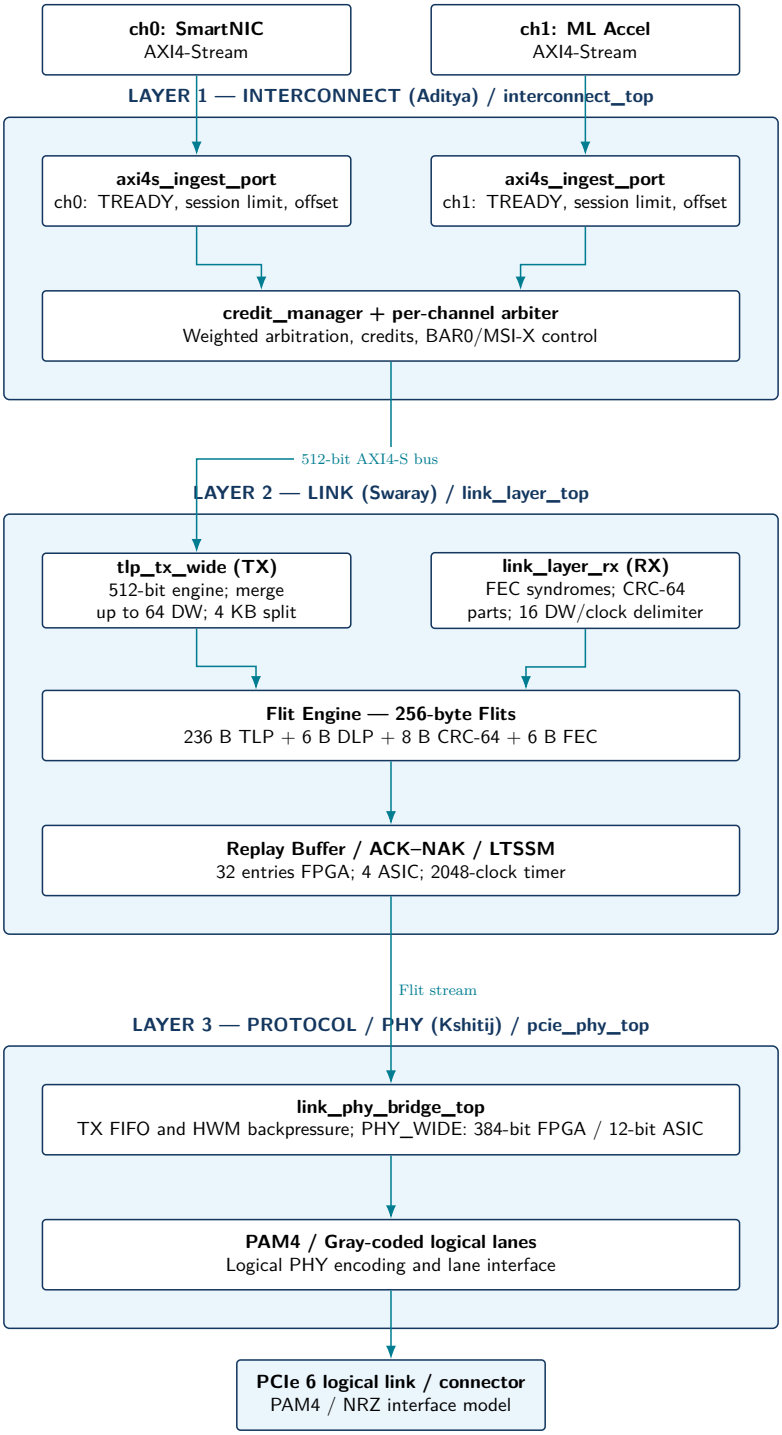


Figure 1: Integrated Round 2 AXI4-Stream to PCIe 6 bridge. TX data flow is emphasized; RX logic is also shown.

3.2 Interconnect layer — Aditya

Two independent 512-bit AXI4-Stream ingress channels accept data using TREADY/TVALID handshakes. Each tracks session activity, a configurable session limit, byte offsets and overflow status. The arbitration and credit manager selects eligible beats; weighted round-robin makes both

SmartNIC and ML priorities programmable. Offset generation is separated from host buffer base address translation: the link-side logic combines a channel's host-programmed base address with its session offset. A Gray-coded FIFO provides clock-domain crossing from AXI-side `aclk` to the PCIe datapath clock. Round 2 corrected sideband bit ordering, session-start crossing, shared backpressure and TKEEP-dependent critical-path behavior (patch X.1).

3.3 Link layer — Swaray

The `tlp_tx_wide` transmit datapath operates on a full 512-bit beat, merges compatible payload segments up to the negotiated maximum payload and splits transfers at 4 KB address boundaries. A 256-byte PCIe 6 Flit carries the documented 236 B TLP area, 6 B DLP, 8 B CRC-64 and 6 B FEC. Receive logic includes parallel FEC syndrome calculation, CRC-64 decomposition and a 16-doubleword-per-clock TLP delimiter. The replay path retains transmitted Flits and processes ACK/NAK, timeouts and link-down recovery. A key Round 2 fix prevented a flow-control DLP from incorrectly consuming a payload Flit. The PHY-ready interface now enforces backpressure, preventing an earlier 32-fold TX/RX rate mismatch.

3.4 Protocol / PHY layer — Kshitij

The `link_phy_bridge_top` connects link-layer Flits to a configurable logical PHY, including a TX FIFO and high-water-mark backpressure. The FPGA-wide profile uses 384-bit, 32-symbol/lane/clock transfer; a reduced 12-bit profile is used for area-constrained SKY130 experiments. Logical Gray/PAM4 mapping, lane handling, configuration, hot reset, BAR0/MSI-X registers and LTSSM recovery are represented at RTL. The verification loopback does *not* constitute a measured external analog PCIe 6.0 serial link.

3.5 System interface and clocks

Interface	Description
AXI ingress (each channel)	512-bit TDATA, 64-bit TKEEP, TLAST, TVALID/TREADY.
Interconnect to link	512-bit TDATA, 64-bit TKEEP, 26-bit TUSER (stream identifier, 24-bit session offset and last indicator, per the design's documented field ordering), valid/ready.
Control	BAR0 configuration, session limits, arbitration weights, MSI-X, errors, reset and link status.
Clocking	FPGA constraints in the run: `aclk` 5.4 ns (185.185 MHz), `clk` 4.0 ns (250 MHz). SKY130 interconnect examples use 30 ns/15 ns constraints.

Table 1: Interfaces and measured/declared design clocks.

4 Integration and Optimisation Work

Phase	Major engineering change	Evidence/status
17–20 Oct	Merge/review independent layers; repair parser, serializer, replay ACK source, TUSER ordering, backpressure, control plane	27 design bugs fixed in review/tickets
20 Oct	Close integration gaps G1–G7; session/translation and host BAR0 control paths	144/144 TLPs in Step 1
21–22 Oct	CI, randomized 40-seed testing, formal, coverage, regression guards	46 guards; 73/73 required bins
22–23 Oct	512-bit wide TX, PHY widening, RX parallelization; CRC/FEC pipelining and XOR tree rewrite	118 to 248 MHz, then 255 MHz in final ledger
23–24 Oct	Workload analyzer and X.1–X.4 ASIC-oriented patches	Performance rerun; PHY/ASIC synthesis fixes

Table 2: Chronological Round 2 development record.

The wide-path optimization initially degraded timing because an unrolled CRC operation created a 51-level combinational path. Partitioning CRC-64 into chunks and expressing CRC/FEC syndromes as constant-mask XOR trees progressively recovered timing. Additional inbound pipelining and TREADY restructuring eliminated the remaining critical paths. RTL changes were checked for behavioral equivalence through the existing directed, random and error-injection benches.

5 Simulation and Testbench Verification

5.1 Test strategy and suites

The integrated `bridge_top` is exercised through a simulated logical PHY loopback, with data integrity checked at RX. Test stimuli cover normal transfers, 4 KB splits, unaligned and partial final beats, arbitration contention, register writes, interrupts, error injection, resets and link-loss recovery. The project uses Icarus Verilog, cocotb, and SymbiYosys formal checks, with GitHub Actions regression automation.

Verification suite	Passing / total	Scope
Interconnect directed benches	4/4	Ingest/arbitration/session
Link directed benches	17/17	TX, RX, framing, replay
Protocol/PHY benches	25/25	Logical lanes, reset, control
Bridge integration benches	3/3	End-to-end TLP and overflow checks
cocotb tests (GitHub final)	10/10	PHY/control simulation
Formal properties	4/4	Gray/PAM4/SOF/header
Guard rules	46/46	Prior regression fixes retained
Random coverage	73/73	40-seed merged required bins

Table 3: Reported final regression outcomes from the Round 2 deliverables ledger (24 October).

The error-injection directed suite reports 19 corrected FEC errors and four replay events with delivered data correct. The bridge directed suite includes 143 TLPs in its main case and 34 TLPs in an overflow scenario. The long-run/nightly suite sends 600 TLPs. Formal properties cover Gray encode/decode inversion, PAM4 mapping, start-of-Flit stripping and TLP header parsing.

5.2 Randomization, coverage and fault recovery

Across 40 randomized seeds the merged coverage report records 1,069 FEC corrections, 433 uncorrectable Flits, 175 CRC failures, 262 replays, 2,197 translation misses, 53 session-limit overflows and 248 MSI-X deliveries. It also reaches 10 hot resets, seven cable pulls, five mid-session signal glitches, straight and reversed lane ordering, and arbitration wins for both sources at weights one through seven. The coverage summary counts *required functional bins*; some optional conditions are covered instead by directed suites. These are simulation/coverage event counts, not measured hardware failure rates.

6 Application-Level Performance Results

An in-house analyzer timestamps stream ingress and reconstructed receive egress to produce throughput, packet rate, percentile latency, FIFO occupancy and recovery statistics. The following numbers are *simulation-derived* at the 250 MHz design-clock normalization, not board-measured PCIe throughput.

Metric / scenario	Initial integrated base	Final	Improvement
ML tensor, 64 KB throughput	0.061 GB/s	6.17 GB/s	∼101×
SmartNIC IMIX rate	0.15 Mpps	4.84 Mpps	32.3×
SmartNIC IMIX p99 latency	58 μs	0.59 μs	98.3× lower
64 B back-to-back rate	0.48 Mpps	11.6 Mpps	24.2×
NIC p99, mixed 1:7 weights	241 μs	4.2 μs	57.4× lower

Table 4: Workload simulation comparison, baseline `step4-base` against final X.1–X.4 source snapshot.

Scenario	Final measured throughput	Interpretation
W1 IMIX SmartNIC	1.72 GB/s	4.84 Mpps packet rate
W2 64 KB ML	6.17 GB/s	p99 1.26 μs
W2 1 MB ML	6.17 GB/s	Sustained bulk traffic
W2 near 4 KB boundary	6.17 GB/s	Split-path stress test
W3 1:1 NIC / ML	0.70 / 5.24 GB/s	Balanced weights
W3 1:3 NIC / ML	0.59 / 5.68 GB/s	Prioritize ML
W3 1:7 NIC / ML	0.43 / 5.69 GB/s	Stronger ML priority
W3 with 300 ppm noise	0.01 / 0.09 GB/s	Replay-limited operation

Table 5: Final workload simulation scenarios.

Heavy logical channel noise sharply reduces throughput because retransmissions dominate. At 2,000–3,000 ppm, certain MSI-X/BAR0 test cases can time out; the project notes no silent data corruption in those runs, but these timeouts remain an open verification/performance issue.

7 FPGA Synthesis, Timing and Power

7.1 Resource utilization (supplied Vivado report)

The attached `synth_utilization.txt` was generated by Vivado 2026.1 for `bridge_top` on xcu250-figd2104-2L-e in the *synthesized* design state. It reports:

Resource	Used	Available	Utilization
CLB LUTs	57,184	1,728,000	3.31%
LUTs as logic	54,044	1,728,000	3.13%
LUTs as distributed RAM	3,140	791,040	0.40%
CLB registers	39,225	3,456,000	1.13%
CARRY8	219	216,000	0.10%
F7 muxes	1,958	864,000	0.23%
F8 muxes	644	432,000	0.15%
Block RAM tiles	0	2,688	0%
URAM	0	1,280	0%

Table 6: Exact synthesized FPGA utilization from the Vivado resource report.

No dedicated block RAM or URAM is inferred in this report; elastic storage uses distributed logic/memory structures. Compared with the team ledger's `step4-base` 56,043 LUTs/24,482 FFs, the final high-throughput datapath has similar LUT scale and more pipeline registers.

7.2 Timing analysis: report provenance matters

The `post_route_timing.txt` is named "post-route," but its header explicitly states *Design State: Synthesized*. It therefore cannot on its own substantiate post-place-and-route timing. For this specific run, clocks are constrained at 5.4 ns (`aclk`) and 4.0 ns (`clk`); the summary reports WNS +0.349 ns, TNS 0, WHS +0.038 ns and zero failing setup/hold endpoints. However, 1,607 input and 921 output ports have no specified input/output delay, so comprehensive interface timing closure has *not* been established by that report alone.

Separate optimization log reports the final X.1–X.4 run as 255 MHz, WNS +0.072 ns and zero failing endpoints; counts in that log (57,245 LUTs/39,269 FFs) differ slightly from the separate synthesis report (57,184/39,225). The two datasets are documented as *distinct reported runs* and are not silently combined.

Development point	LUTs	FFs	WNS at 4 ns	Failing endpoints
Step 4 base	56,043	24,482	−4.439 ns	19,046
Wave A+B	67,999	36,244	−16.285 ns	not reported
Wave C	61,914	39,147	−2.789 ns	18,164
Freeze	60,816	39,304	−0.027 ns	82
Final X.1–X.4 (team ledger)	57,245	39,269	+0.072 ns	0

Table 7: Vivado optimization history ; independent of the separately attached timing report.

7.3 Power estimate

The Vivado power estimate for the synthesized xcu250 design gives 3.499 W total on-chip power, consisting of 0.545 W dynamic and 2.954 W device static power (medium confidence). Dynamic components include approximately 0.273 W clocks, 0.155 W CLB logic and 0.118 W signals. No simulation activity file is matched, so this should be treated as a tool estimate rather than measured operating power. Its FPGA device-static component must not be mistaken for SKY130 ASIC power.

8 ASIC Physical Implementation and GDSII Status

The physical-design exploration uses LibreLane/OpenROAD with the SkyWater SKY130A `sky130_fd_sc_hd` library. Clock-domain assumptions and reduced buffer/PHY parameters differ from the high-speed

FPGA target. The `interconnect_top` run completed GDSII stream-out with DRC, LVS and antenna checks reported clean, and hold timing met. The early run reports 1.20 mm^2 die area, 41% utilization, about 49,000 cells (9,800 FFs), 104 mW reported power and a slow-corner setup violation of -0.95 ns; X.1 improved its setup slack to -0.52 ns, but that follow-up run was stopped before a new GDSII stream-out due to time-constraints.

The separate `link_layer_rx` synthesis check passes after correction-logic changes X.3/X.4; its reported standard-cell area is approximately 1.12 mm^2. A full-bridge narrow-profile synthesis estimate is roughly 3.9 mm^2 of standard cells; this is *not* a verified final die footprint. At the documented 24 October cutoff the full-bridge flows (narrow 30 ns, conservative narrow 50 ns and larger wide build) were still in progress. Thus, no completed, DRC/LVS-clean *full-bridge* GDSII is asserted here.

8.1 GDSII Physical Layout Visualization

Figures 2 and 3 show the physical layout visualizations obtained from the SKY130 ASIC implementation flow.

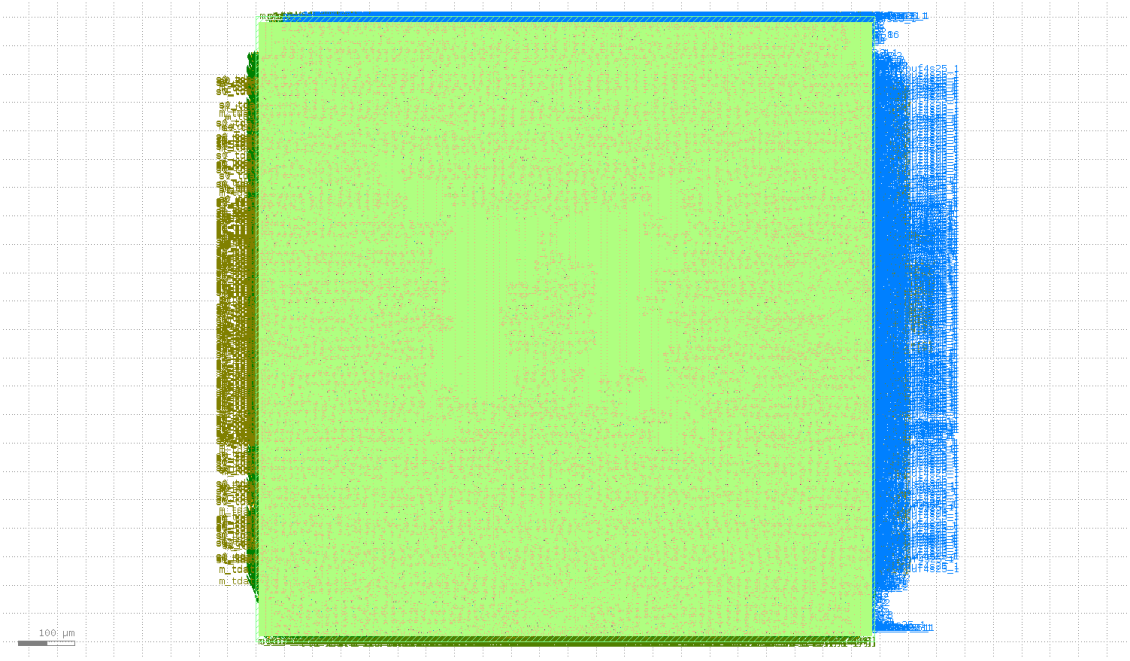


Figure 2: Physical layout overview showing the chip boundary, routing structures, and peripheral signal connections.

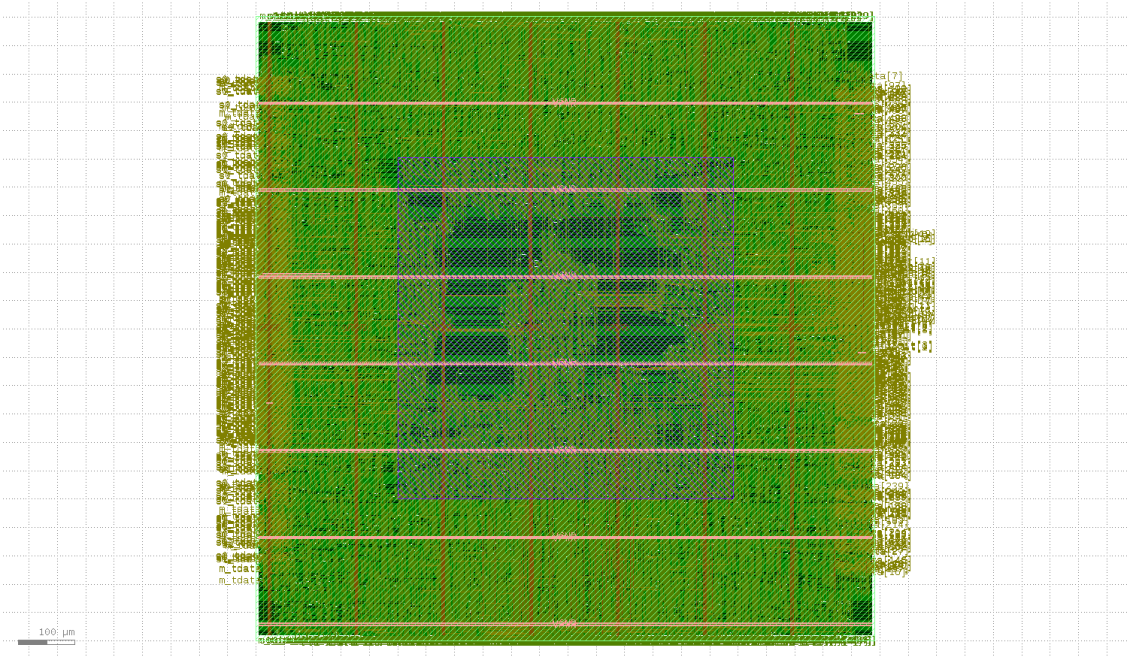


Figure 3: Alternative physical layout view highlighting interconnect routing layers, metal structures, and power distribution geometry.

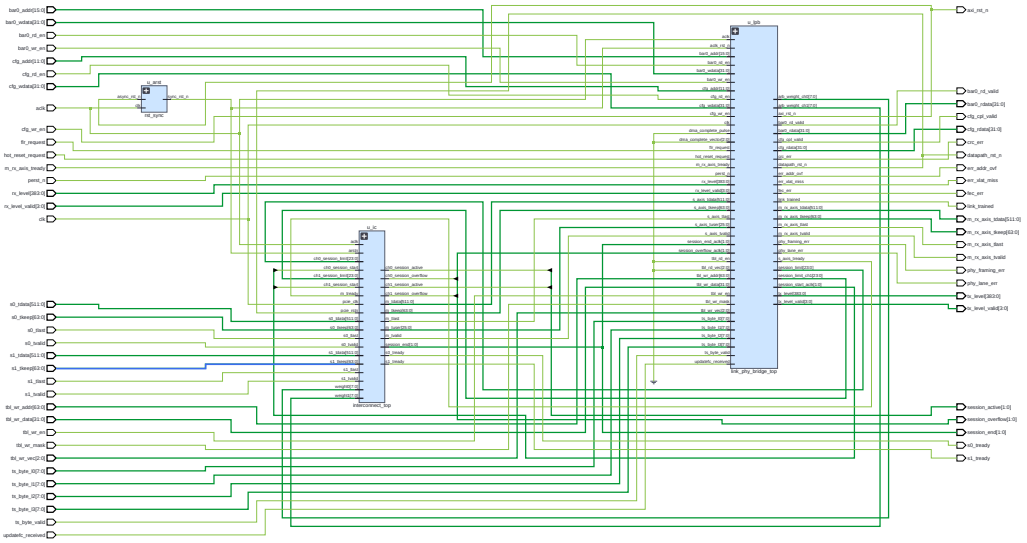
9 Synthesis Schematics and Supporting Figures

The appended schematics are the actual user-supplied design outputs, reproduced as vector PDF pages: an elaborated-design view, synthesized-design view and implemented-design view. Their presence demonstrates available design views but does not change the synthesized-state timing report's provenance. The diagrams contain dense RTL/netlist connectivity and are provided at full page scale to permit zooming.

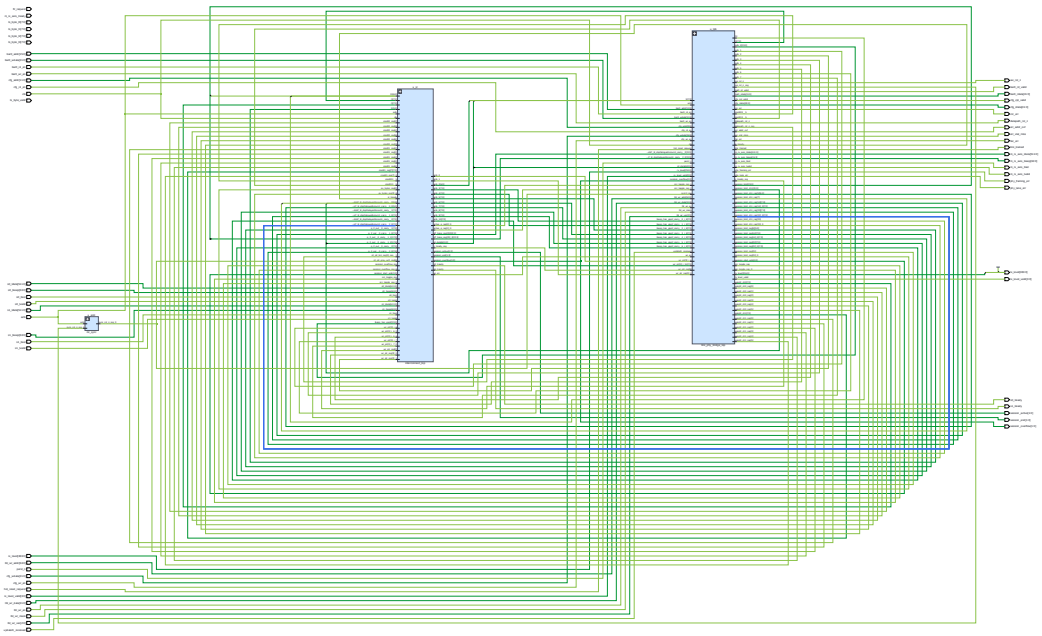
10 Limitations and Conclusion

Round 2 delivered integrated RTL, wide datapath optimizations, cross-layer flow control, robust regression infrastructure, 40-seed functional coverage and simulation improvements for both target workloads. FPGA resource utilization is low relative to the selected xcu250, and the supplied 250 MHz-constrained timing summary has positive internal timing margins. Remaining qualifications are important: the simulation PHY is not a characterized physical PCIe 6.0 SerDes; the timing text describes a synthesized-state design despite its filename; several I/O delays are unconstrained; heavy channel-noise throughput is poor; and only the interconnect subblock has a documented completed, clean SKY130 GDSII at the provided cutoff. Completing and independently checking the full-bridge route, corners and manufacturability reports is the next milestone.

A Elaborated RTL Design Schematic



B Synthesized Design Schematic



C Implemented Design Schematic

